My Climate Story

Learning Climate Change Through Your Own Backyard

The Problem

Climate change is often taught as a global, abstract topic: rising global temperatures, melting glaciers, and average sea levels. For students in climate-vulnerable regions like Bangladesh, this feels distant from what they actually see around them, such as shifting rivers, hotter summers, and irregular monsoons. Most climate education material is also in English, data-heavy, and assumes fast internet and a technical background. As a result, the young people who will live with the consequences the longest have the fewest tools to understand what is changing in their own community.

Our Solution

My Climate Story is a lightweight, mobile-friendly web app that lets a student pick any location on the map and see how their own area has changed over the past decades, using NASA satellite imagery and climate data. Instead of showing global averages, it tells a local, personal story in simple language.

What the App Does

1. **Choose your place.** Users select their village, town, or district on an interactive map.
2. **See the change.** A before-and-after slider compares satellite imagery of the same place across different years, showing river movement, urban growth, or vegetation loss.
3. **Understand the trend.** Simple charts show how temperature and rainfall at that exact location have changed since the 1980s, with a one-line plain-language explanation (for example, "Your area now has more hot days than in your grandparents' time").
4. **Learn by playing.** A short interactive quiz and guessing game (such as "Which image is from 2000?") makes learning active rather than passive.
5. **Take action.** A "What can I do?" section gives practical, local steps students and schools can take.

NASA Data We Use

- **NASA GIBS / Worldview:** historical satellite imagery for the before-and-after comparison
- **NASA POWER:** location-specific temperature and rainfall records since 1981
- **NASA GISTEMP and the Global Climate Change portal:** global context to connect local change to the bigger picture
- **(Optional) NASA Sea Level Change Portal:** for coastal communities

Why It Is Different

- **Hyperlocal:** it shows "your area," not a global average.
- **Language accessible:** built first in Bangla, with a language switcher so other communities can adopt it (English, Hindi, Swahili, and more).
- **Designed for low connectivity:** lightweight pages and cached data so it works in places with weak internet.

- **Storytelling-first:** data is presented as a narrative that students can relate to, not as raw charts.

Impact and Scalability

Our primary users are school students (roughly grades 6 to 10) and teachers. Because the app relies only on global, open NASA datasets, it works for any location on Earth. A student in the Mekong Delta, the Nile Delta, or the Pacific Islands can use it in exactly the same way, and adding a new language requires only translated text. Teachers can use it as a ready classroom activity with no special equipment.

Technology

- **Frontend:** HTML/JavaScript (or React), Leaflet for maps, Chart.js for graphs
- **Data:** NASA GIBS and NASA POWER APIs, called directly from the browser
- **Hosting:** GitHub Pages or Netlify (free)
- **Design:** Figma for the initial prototype

Future Work

- Teacher dashboard and lesson plans
- Community photo contributions to compare with satellite views
- Future climate projections using NASA NEX-GDDP-CMIP6 data
- Offline mode and SMS/WhatsApp-based summaries for very low-connectivity areas